

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes

AIMLC ZC416

Mathematical Foundations for Machine Learning

COMPREHENSIVE SELF-STUDY NOTES

Contact Hours 1 – 8 | Mid-Semester Examination

Concepts • Theorems • Worked Examples • Problem-Solving Techniques

Prepared for: Intermediate-level students targeting exam readiness

SYLLABUS OVERVIEW: Hours 1–8

This document covers all topics in the Mid-Semester examination syllabus. Each section is structured with: core concept explanation, key formulas, worked examples, and exam problem-solving strategies.

Hr	Topic	Key Sub-Topics	Ref
1	Solution of Linear Systems	Matrices, Gaussian Elim, RREF, Rank, Consistency, Inverse	T1: 2.1-2.3
2	Vector Spaces	Linear Independence, Basis, Dimension, Rank-Nullity, Null/Column Space	T1: 2.4-2.6,2.8
3	Analytic Geometry	Norms, Inner Products, Cauchy-Schwarz, Projection, Gram-Schmidt, Orthonormal Basis	T1: 3.1-3.5
4	Matrix Decomposition I	Determinant, Trace, Eigenvalues & Eigenvectors, Cholesky	T1: 4.1-4.3
5	Matrix Decomposition II	Eigendecomposition (EVD), SVD, Low-Rank Approximation	T1: 4.4-4.6
6	Vector Calculus I	Univariate derivatives, Partial derivatives, Gradient, Jacobian	T1: 5.1-5.3
7	Vector Calculus II	Matrix Gradients, Useful Identities, Backpropagation	T1: 5.4-5.6
8	Vector Calculus III	Higher-order derivatives, Hessian, Taylor Series, Unconstrained Optimization	T1: 5.7-5.8

HOUR 1: Solution of Linear Systems

1.1 What is a System of Linear Equations?

A system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ has the form:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

In matrix form: $Ax = b$, where A is $m \times n$, x is $n \times 1$, b is $m \times 1$.

NOTE The matrix A holds the coefficients. x holds the unknowns. b holds the right-hand side values. This compact notation lets us use powerful linear algebra tools.

1.2 Types of Solutions

Type	Condition	Geometric Meaning (2D)
Unique Solution	$\text{rank}(A) = \text{rank}([A b]) = n$	Lines intersect at exactly one point
No Solution	$\text{rank}(A) < \text{rank}([A b])$	Lines are parallel (inconsistent)
Infinitely Many	$\text{rank}(A) = \text{rank}([A b]) < n$	Lines are the same (overlapping)

1.3 Gaussian Elimination – The Core Algorithm

Gaussian Elimination converts the augmented matrix $[A|b]$ to Row Echelon Form (REF) using three elementary row operations (EROs):

- Row Swap: $R^i \leftrightarrow R_j$
- Row Scale: $R^i \rightarrow k \cdot R^i$ ($k \neq 0$)
- Row Combination: $R^i \rightarrow R^i + k \cdot R_j$

Row Echelon Form (REF) vs Reduced Row Echelon Form (RREF)

Property	REF	RREF
All-zero rows	At the bottom	At the bottom
Leading entry (pivot)	Any nonzero value	Must be 1
Pivots shift right	Yes (each row)	Yes
Column above pivot	Can be nonzero	Must be zero
Unique?	No	Yes (unique for any matrix)

Worked Example 1.1: Solve the system by Gaussian Elimination

System: $x + y + z = 6$
 $2x - y + z = 3$
 $x + 2y - z = 2$

Augmented matrix:	Step 1: Form $[A b]$
$[1 \quad 1 \quad 1 \mid 6]$	
$[2 \quad -1 \quad 1 \mid 3]$	
$[1 \quad 2 \quad -1 \mid 2]$	
R2 $\rightarrow$ R2 - 2*R1:	Step 2: Eliminate below first pivot
$[1 \quad 1 \quad 1 \mid 6]$	
$[0 \quad -3 \quad -1 \mid -9]$	
$[0 \quad 1 \quad -2 \mid -4]$	
R3 $\rightarrow$ R3 + (1/3)*R2:	Step 3: Eliminate below second pivot
$[1 \quad 1 \quad 1 \mid 6]$	
$[0 \quad -3 \quad -1 \mid -9]$	
$[0 \quad 0 \quad -7/3 \mid -7]$	
From row 3: $z = (-7) / (-7/3) = 3$	
From row 2: $-3y - 1(3) = -9 \Rightarrow y = 2$	
From row 1: $x + 2 + 3 = 6 \Rightarrow x = 1$	
$\Rightarrow$ Solution: $x=1, y=2, z=3$ (unique solution)	

1.4 Rank of a Matrix

The rank of a matrix A is the number of pivot columns (leading 1s in RREF). It equals the dimension of the column space.

Key rank property:

$$\text{rank}(A) = \text{rank}(A^T) \quad [\text{row rank} = \text{column rank}]$$

Consistency condition:

$$Ax = b \text{ is consistent IFF } \text{rank}(A) = \text{rank}([A|b])$$

Unique solution condition:

$$\text{Unique solution exists IFF } \text{rank}(A) = \text{rank}([A|b]) = n$$

1.5 Matrix Inverse

An $n \times n$ matrix A is invertible (nonsingular) if and only if $\det(A) \neq 0$. If A is invertible, the unique solution to $Ax = b$ is $x = A^{-1}b$.

2x2 Inverse Formula (Memorize this!):

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow A^{-1} = (1/(ad-bc)) * \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Inverse via Augmented Matrix $[A|I]$:

$$\text{Row-reduce } [A|I] \rightarrow [I|A^{-1}] \quad (\text{if } A \text{ is invertible})$$

Worked Example 1.2: Find inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$

$$\det(A) = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1$$

Since $\det \neq 0$, A is invertible.

$$A^{-1} = (1/1) \cdot \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

$$\begin{aligned} \text{Verify: } A \cdot A^{-1} &= \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \cdot \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \checkmark \end{aligned}$$

1.6 Condition Number

Condition Number:

$$\text{cond}(A) = \|A\| \cdot \|A^{-1}\| \quad (\text{or } \sigma_{\max} / \sigma_{\min} \text{ using SVD})$$

Measures how sensitive the solution x is to small changes in b . A large condition number means the system is ill-conditioned.

- $\text{cond}(A)$ close to 1: well-conditioned (numerically safe)
- $\text{cond}(A)$ very large (e.g., 10^8): ill-conditioned (small errors in b cause huge errors in x)

NOTE In the lab, you solve $Ax=b$ for different A matrices and study how $\text{cond}(A)$ affects accuracy. Matrices near singularity have huge condition numbers.

EXAM TIP If asked 'is this system well-conditioned?', compute or compare singular values: small $\sigma_{\min}$ relative to $\sigma_{\max} \Rightarrow$ ill-conditioned.

HOUR 2: Vector Spaces, Linear Independence, Basis & Rank

2.1 Vector Space – Definition & Axioms

A vector space V over a field F (usually the real numbers $\mathbb{R}$) is a set with two operations: vector addition (+) and scalar multiplication ($\cdot$), satisfying 8 axioms:

Axiom	Requirement
A1: Closure under +	$u, v \in V \Rightarrow u+v \in V$
A2: Commutativity	$u+v = v+u$
A3: Associativity	$(u+v)+w = u+(v+w)$
A4: Zero element	$\exists 0 \in V$ such that $v+0 = v$
A5: Additive inverse	$\forall v, \exists -v$ such that $v+(-v)=0$
A6: Closure under scalar mult	$c \in F, v \in V \Rightarrow c \cdot v \in V$
A7: Distributive (scalar)	$c(u+v) = cu + cv$
A8: Distributive (vector)	$(c+d)v = cv + dv$

NOTE You don't always need to check all 8 axioms. For a subset of a known vector space, use the Subspace Test (3 conditions).

2.2 Subspaces

Subspace Test (3 conditions only)

A nonempty subset S of a vector space V is a subspace if and only if:

1. The zero vector 0 is in S
2. S is closed under addition: if u, v in S , then $u+v$ in S
3. S is closed under scalar multiplication: if v in S and c is a scalar, then cv in S

Worked Example 2.1: Is $S = \{[x,y,z] : x + y + z = 0\}$ a subspace of $\mathbb{R}^3$?

Check 1 - Zero vector: $[0,0,0] : 0+0+0 = 0 \checkmark$

Check 2 - Closure under +:

Let $u=[a,b,c]$ and $v=[d,e,f]$ both satisfy $x+y+z=0$.

$$u+v = [a+d, b+e, c+f]$$

$$(a+d)+(b+e)+(c+f) = (a+b+c)+(d+e+f) = 0+0 = 0 \checkmark$$

Check 3 - Closure under scalar mult:

$$k \cdot u = [ka, kb, kc]$$

$$ka+kb+kc = k(a+b+c) = k \cdot 0 = 0 \checkmark$$

$\Rightarrow S$ is a subspace of $\mathbb{R}^3$. (It is a plane through the origin.)

Worked Example 2.2: Is $T = \{[x,y] : x + y = 1\}$ a subspace of $\mathbb{R}^2$?

Check 1 - Zero vector: $0 + 0 = 0 \neq 1 \Rightarrow$ zero vector NOT in T

$\Rightarrow T$ is NOT a subspace. (It is a line NOT through the origin.)

2.3 Linear Independence

A set of vectors $\{v_1, v_2, \dots, v_k\}$ is linearly independent (LI) if the only solution to:

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k = 0 \text{ is } c_1 = c_2 = \dots = c_k = 0$$

If any nonzero c_i exists, the set is linearly dependent (LD). A dependent set means one vector can be written as a linear combination of the others.

How to Check Linear Independence

1. Form the matrix M whose columns are the given vectors.
2. Row-reduce M to REF/RREF.
3. If every column has a pivot: the vectors are linearly independent.
4. If any column is free (no pivot): the vectors are linearly dependent.

Worked Example 2.3: Are $v_1=[1,2,3]$, $v_2=[2,4,6]$, $v_3=[1,0,1]$ linearly independent?
Form matrix with columns v_1, v_2, v_3 :
$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 3 & 6 & 1 \end{bmatrix}$
$R_2 \rightarrow R_2 - 2 \cdot R_1$:
$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix}$
$R_3 \rightarrow R_3 - R_2$:
$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$
Only 2 pivots for 3 columns $\Rightarrow$ column 2 is FREE $\Rightarrow v_2$ is dependent.
Indeed: $v_2 = 2 \cdot v_1$ (v_2 is just a scalar multiple of v_1).
$\Rightarrow$ The set is LINEARLY DEPENDENT.

2.4 Span, Basis, Dimension

Term	Definition	Key Fact
Span	Set of all linear combinations of vectors	$\text{span}\{v_1, \dots, v_k\}$ is always a subspace
Basis	A LI set that spans the entire space V	Not unique; size is unique (= dimension)

Term	Definition	Key Fact
Dimension	Number of vectors in any basis of V	$\dim(\mathbb{R}^n) = n$; $\dim(\text{null space}) = \text{nullity}$
Coordinates	Representation in terms of basis vectors	Unique for a given basis

2.5 Rank-Nullity Theorem

Rank-Nullity Theorem (FUNDAMENTAL - memorize!)

For any $m \times n$ matrix A : $\text{rank}(A) + \text{nullity}(A) = n$ where $\text{rank}(A) = \text{number of pivots} = \dim(\text{column space})$ and $\text{nullity}(A) = \dim(\text{null space}) = \text{number of free variables}$

Worked Example 2.4: Find basis for null space of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$

First row-reduce A :

$R_2 \rightarrow R_2 - 2R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Pivots: column 1 (x_1 is pivot variable)

Free variables: x_2, x_3

$\text{rank}(A) = 1, \quad \text{nullity} = 3 - 1 = 2$

From row 1: $x_1 + 2x_2 + 3x_3 = 0 \Rightarrow x_1 = -2x_2 - 3x_3$

Let $x_2 = s, x_3 = t$ (free):

$$\mathbf{x} = [-2s - 3t, s, t] = s[-2, 1, 0] + t[-3, 0, 1]$$

=> Basis for Null Space: $\{ [-2, 1, 0], [-3, 0, 1] \}$

2.6 The Four Fundamental Subspaces

Subspace	How to Find	Dimension
Column Space $C(A)$	Pivot columns of A (original, not RREF)	$\text{rank}(A)$
Null Space $N(A)$	Solve $A\mathbf{x} = \mathbf{0}$; write parametric solution	$n - \text{rank}(A)$
Row Space $C(A^T)$	Non-zero rows of REF of A	$\text{rank}(A)$
Left Null Space $N(A^T)$	Solve $A^T \mathbf{y} = \mathbf{0}$	$m - \text{rank}(A)$

EXAM TIP Exam shortcut: If $\text{rank}(A) = r$ for an $m \times n$ matrix: $\dim(C(A)) = r$, $\dim(N(A)) = n - r$, $\dim(C(A^T)) = r$, $\dim(N(A^T)) = m - r$. These must add up correctly.

HOUR 3: Analytic Geometry – Norms, Inner Products, Orthogonality

3.1 Norms

A norm $\|\cdot\|$ measures the 'length' or 'size' of a vector. It must satisfy:

- Non-negativity: $\|x\| \geq 0$, and $\|x\|=0$ iff $x=0$
- Absolute homogeneity: $\|cx\| = |c| \cdot \|x\|$
- Triangle inequality: $\|x+y\| \leq \|x\| + \|y\|$

Norm	Formula	Notes / ML Usage
L1 (Manhattan)	$\ x\ _1 = x_1 + x_2 + \dots + x_n $	Promotes sparsity (Lasso regularization)
L2 (Euclidean)	$\ x\ _2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$	Default norm; Ridge regularization ($\ x\ _2^2$)
Lp (General)	$\ x\ _p = (\sum x_i ^p)^{1/p}$	Generalizes all norms for $p \geq 1$
L ∞ (Max)	$\ x\ _\infty = \max(x_i)$	Robust; Chebyshev distance
Frobenius (Matrix)	$\ A\ _F = \sqrt{\sum_{ij} a_{ij}^2}$	$= \sqrt{\text{tr}(A^T A)} = \sqrt{\sum \sigma_i^2}$

Worked Example 3.1: Compute all norms of $x = [3, -4]$

L1: $\|x\|_1 = |3| + |-4| = 3 + 4 = 7$

L2: $\|x\|_2 = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$

L-inf: $\|x\|_\infty = \max(3, 4) = 4$

Note: L2 gives the familiar Euclidean length (Pythagorean theorem).

3.2 Inner Products

The inner product is a generalization of the dot product. The standard inner product on $\mathbb{R}^n$ is:

Standard Inner Product (dot product):

$$\langle x, y \rangle = x^T y = \sum (x_i \cdot y_i) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

A general inner product must satisfy: symmetry, linearity, positive definiteness.

Key relationship:

$$\|x\|_2^2 = \langle x, x \rangle = x^T x$$

3.3 Geometric Properties

Cauchy-Schwarz Inequality

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

Equality holds if and only if x and y are parallel (one is a scalar multiple of the other).

Angle Between Vectors

Angle formula:

$$\cos(\theta) = \langle \mathbf{x}, \mathbf{y} \rangle / (||\mathbf{x}|| \cdot ||\mathbf{y}||)$$

- If $\langle \mathbf{x}, \mathbf{y} \rangle = 0$: vectors are orthogonal ($\theta = 90^\circ$)
- If $\langle \mathbf{x}, \mathbf{y} \rangle > 0$: angle is acute ($\theta < 90^\circ$)
- If $\langle \mathbf{x}, \mathbf{y} \rangle < 0$: angle is obtuse ($\theta > 90^\circ$)

Projection of x onto y

Projection formula:

$$\text{proj}_{\mathbf{y}}(\mathbf{x}) = (\langle \mathbf{x}, \mathbf{y} \rangle / \langle \mathbf{y}, \mathbf{y} \rangle) \cdot \mathbf{y} = (\langle \mathbf{x}, \mathbf{y} \rangle / ||\mathbf{y}||^2) \cdot \mathbf{y}$$

Projection onto unit vector u:

$$\text{proj}_{\mathbf{u}}(\mathbf{x}) = \langle \mathbf{x}, \mathbf{u} \rangle \cdot \mathbf{u}$$

Distance from x to line spanned by y:

$$d = ||\mathbf{x} - \text{proj}_{\mathbf{y}}(\mathbf{x})||$$

Worked Example 3.2: Project $\mathbf{x}=[3,1]$ onto $\mathbf{y}=[2,2]$

$$\langle \mathbf{x}, \mathbf{y} \rangle = 3 \cdot 2 + 1 \cdot 2 = 8$$

$$\langle \mathbf{y}, \mathbf{y} \rangle = 2 \cdot 2 + 2 \cdot 2 = 8$$

$$\text{proj}_{\mathbf{y}}(\mathbf{x}) = (8/8) \cdot [2, 2] = 1 \cdot [2, 2] = [2, 2]$$

$$\text{Error vector (perpendicular): } \mathbf{x} - \text{proj} = [3, 1] - [2, 2] = [1, -1]$$

$$\text{Check: } \langle [1, -1], [2, 2] \rangle = 2 - 2 = 0 \quad \checkmark \quad (\text{perpendicular confirmed})$$

3.4 Orthogonality and Orthonormal Bases

Term	Definition	Property
Orthogonal set	$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $i \neq j$	Any orthogonal set of nonzero vectors is LI
Orthonormal set	Orthogonal AND $ \mathbf{v}_i =1$ for all i	Simplifies projections: $\text{proj} = \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i$
Orthogonal matrix Q	$Q^T Q = I$ (square)	$Q^{-1} = Q^T$; preserves lengths and angles

3.5 Gram-Schmidt Orthogonalization

Converts any basis $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ into an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ spanning the same space.

Algorithm (Step-by-Step)

5. Set $\mathbf{v}'_1 = \mathbf{v}_1$
6. For each $k = 2, 3, \dots, n$: subtract the projection onto all previous orthonormal vectors:

$$\mathbf{v}'_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{v}_k, \mathbf{u}_j \rangle \mathbf{u}_j$$

7. Normalize each vector:

$$\mathbf{u}_k = \mathbf{v}'_k / ||\mathbf{v}'_k||$$

Worked Example 3.3: Apply Gram-Schmidt to $v_1=[1,1,0]$, $v_2=[1,0,1]$, $v_3=[0,1,1]$

Step 1: $u_1 = v_1 / ||v_1||$

$$||v_1|| = \sqrt{1+1+0} = \sqrt{2}$$

$$u_1 = [1/\sqrt{2}, 1/\sqrt{2}, 0]$$

Step 2: Find $v'_2 = v_2 - \langle v_2, u_1 \rangle u_1$

$$\langle v_2, u_1 \rangle = 1 \cdot (1/\sqrt{2}) + 0 \cdot (1/\sqrt{2}) + 1 \cdot 0 = 1/\sqrt{2}$$

$$v'_2 = [1, 0, 1] - (1/\sqrt{2}) \cdot [1/\sqrt{2}, 1/\sqrt{2}, 0]$$

$$= [1, 0, 1] - [1/2, 1/2, 0] = [1/2, -1/2, 1]$$

$$||v'_2|| = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2}$$

$$u_2 = [1/2, -1/2, 1] / \sqrt{3/2}$$

Step 3: Similarly find v'_3 by removing projections onto u_1 and u_2 ...

(Process continues until all vectors are orthonormalized)

=> Result: orthonormal set $\{u_1, u_2, u_3\}$ spanning same subspace as $\{v_1, v_2, v_3\}$

EXAM TIP Gram-Schmidt is a lab topic. Know the formula for the orthogonal component: $v'_k = v_k$ minus the sum of its projections onto all previous orthonormal vectors. Always normalize at the end.

HOURL 4: Matrix Decomposition I – Determinant, Trace, Eigenvalues, Cholesky

4.1 Determinant

The determinant $\det(A)$ is a scalar value associated with a square matrix. It has deep geometric meaning: $|\det(A)|$ = volume of the parallelepiped formed by the rows (or columns) of A .

Computation Formulas

2x2 Determinant:

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3x3 by Cofactor Expansion along row 1:

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \text{ where cofactor } C_{ij} = (-1)^{i+j} * M_{ij} \\ (M_{ij} = \text{minor} = \det \text{ of } (n-1) \times (n-1) \text{ submatrix})$$

Triangular Matrix (Key shortcut!):

$$\det(\text{upper or lower triangular } A) = a_{11} * a_{22} * a_{33} * \dots * a_{nn} \text{ (product of diagonal)}$$

Key Properties of Determinant

Property	Formula
Product rule	$\det(AB) = \det(A) \cdot \det(B)$
Transpose	$\det(A^T) = \det(A)$
Inverse	$\det(A^{-1}) = 1/\det(A)$
Scalar multiple	$\det(cA) = c^n \cdot \det(A)$ [$n \times n$ matrix]
Row swap	Swapping 2 rows multiplies det by -1
Row scale	Multiplying row by k multiplies det by k
Row combination	Adding $k \cdot (\text{row } i)$ to row j : det unchanged
Singular matrix	$\det(A) = 0$ iff A is not invertible

Worked Example 4.1: Compute $\det\begin{pmatrix} 2 & 1 & 0 \\ 3 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$

Cofactor expansion along row 1:

$$\det = 2 * \det\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} - 1 * \det\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} + 0 * (\text{anything})$$

$$\det\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} = 3*2 - 1*1 = 5$$

$$\det\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = 1*2 - 1*0 = 2$$

$$\det(A) = 2*5 - 1*2 + 0 = 10 - 2 = 8$$

4.2 Trace

Trace:

$$\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn} = \text{sum of diagonal entries} = \text{sum of all eigenvalues}$$

Property	Formula
Cyclic	$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$ [order is cyclic, not arbitrary]
Linearity	$\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$; $\text{tr}(cA) = c \cdot \text{tr}(A)$
Transpose	$\text{tr}(A^T) = \text{tr}(A)$
Eigenvalues	$\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$

4.3 Eigenvalues and Eigenvectors

Core Definition

A scalar λ is an eigenvalue of A , and nonzero vector v is its eigenvector, if: $A \cdot v = \lambda \cdot v$
Interpretation: A only scales v (no rotation). λ is the scale factor.

Step-by-Step: Finding Eigenvalues and Eigenvectors

8. Solve the characteristic equation: $\det(A - \lambda I) = 0$
9. This gives a polynomial in λ (degree n for $n \times n$ matrix). Solve for all roots $\lambda_1, \lambda_2, \dots$
10. For each eigenvalue λ_i : form the matrix $(A - \lambda_i I)$
11. Solve $(A - \lambda_i I)v = 0$ by row reduction; express solution in parametric form
12. The parametric vectors are the eigenvectors (forming the eigenspace for λ_i)

Worked Example 4.2: Find eigenvalues & eigenvectors of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

STEP 1: Characteristic equation

$$\begin{aligned}
 \det(A - \lambda I) &= \det(\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}) \\
 &= (4-\lambda)(3-\lambda) - (1)(2) \\
 &= 12 - 7\lambda + \lambda^2 - 2 \\
 &= \lambda^2 - 7\lambda + 10 = 0
 \end{aligned}$$

STEP 2: Solve: $(\lambda - 5)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 5, \lambda_2 = 2$

STEP 3: Verify: $\text{tr}(A) = 4+3 = 7 = 5+2 \checkmark \quad | \quad \det(A) = 12-2 = 10 = 5 \cdot 2 \checkmark$

STEP 4a: Eigenvector for $\lambda_1 = 5$:

$$(A - 5I) = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Row reduce: $R_2 \rightarrow R_2 + 2R_1$: $\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow -v_1 + v_2 = 0 \Rightarrow v_1 = v_2$

Eigenvector: $v = [1, 1]$ (or any nonzero scalar multiple)

STEP 4b: Eigenvector for $\lambda_2 = 2$:

$$(A - 2I) = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

Row reduce: $R_2 \rightarrow R_2 - R_1$: $\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow 2v_1 + v_2 = 0 \Rightarrow v_2 = -2v_1$

Eigenvector: $v = [1, -2]$

STEP 5: Verify $A \cdot [1,1] = [4+1, 2+3] = [5,5] = 5 \cdot [1,1] \checkmark$

Important Properties of Eigenvalues

Property	Detail
Trace = sum of EVs	$\text{tr}(A) = \lambda_1 + \dots + \lambda_n$
Det = product of EVs	$\det(A) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$
Singular matrix	A is singular iff at least one $\lambda = 0$
Symmetric matrix	All eigenvalues are real; eigenvectors orthogonal
A^k eigenvalues	If $Av = \lambda v$, then $A^k v = \lambda^k v$
Invertible eigenvalues	If $Av = \lambda v$ and $\lambda \neq 0$, then $A^{-1}v = (1/\lambda)v$

4.4 Cholesky Decomposition

Every symmetric positive definite (SPD) matrix A can be factored as $A = LL^T$, where L is a lower triangular matrix with positive diagonal entries.

Positive Definite (PD) Matrix – Check

- All eigenvalues $\lambda_i > 0$
- $x^T A x > 0$ for all nonzero x
- All leading principal minors are positive (Sylvester's criterion)

Cholesky Algorithm for 3×3

Diagonal entries:

$$L_{jj} = \sqrt{A_{jj} - \sum_{k=1}^{j-1} L_{jk}^2}$$

Off-diagonal entries ($i > j$):

$$L_{ij} = (A_{ij} - \sum_{k=1}^{j-1} L_{ik} \cdot L_{jk}) / L_{jj}$$

Worked Example 4.3: Cholesky of $A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$

Check PD: eigenvalues of $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 4 = \lambda^2 - 7\lambda + 8$$

$$\lambda = (7 \pm \sqrt{49 - 32}) / 2 = (7 \pm \sqrt{17}) / 2 \Rightarrow \text{both positive } \checkmark$$

Cholesky: $A = L \cdot L^T$ where $L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$

$$l_{11} = \sqrt{A_{11}} = \sqrt{4} = 2$$

$$l_{21} = A_{21} / l_{11} = 2 / 2 = 1$$

$$l_{22} = \sqrt{A_{22} - l_{21}^2} = \sqrt{3 - 1} = \sqrt{2}$$

$$L = \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{bmatrix}$$

$$\begin{aligned}\text{Verify: } L \cdot L^T &= \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ 2 & 1+2 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} \quad \checkmark\end{aligned}$$

EXAM TIP Cholesky is used to solve $Ax=b$ efficiently: (1) Solve $Ly=b$ by forward substitution, (2) Solve $L^T x=y$ by back substitution. This is faster than full matrix inversion.

HOUR 5: Matrix Decomposition II – EVD, SVD, Low-Rank Approximation

5.1 Eigendecomposition (EVD / Diagonalization)

A square matrix A ($n \times n$) is diagonalizable if it has n linearly independent eigenvectors. Then:

Eigendecomposition:

$$A = P D P^{-1}$$

where P is the $n \times n$ matrix of eigenvectors (as columns) and $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$.

For symmetric A : eigenvectors are orthogonal, so $P = Q$ (orthogonal), giving:

Spectral Decomposition (symmetric A):

$$A = Q D Q^T \quad \text{where } Q^T Q = I \quad (Q^{-1} = Q^T)$$

Power of a Matrix Using EVD

Matrix power shortcut:

$$A^k = P D^k P^{-1}$$

$$D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k) \quad [\text{just raise each diagonal entry to the } k\text{-th power}]$$

Worked Example 5.1: Compute A^5 where $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ (EVs from Hour 4)

From Hour 4: $\lambda_1=5$, $v_1=[1,1]$; $\lambda_2=2$, $v_2=[1,-2]$

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

$$P^{-1}: \det(P) = -2-1 = -3$$

$$P^{-1} = (1/-3) * \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

$$A^5 = P * D^5 * P^{-1}$$

$$D^5 = \begin{bmatrix} 5^5 & 0 \\ 0 & 2^5 \end{bmatrix} = \begin{bmatrix} 3125 & 0 \\ 0 & 32 \end{bmatrix}$$

$$\begin{aligned} A^5 &= \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} * \begin{bmatrix} 3125 & 0 \\ 0 & 32 \end{bmatrix} * \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} \\ &= \begin{bmatrix} 3125 & 32 \\ 3125 & -64 \end{bmatrix} * \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{Final: } A^5 &= \begin{bmatrix} (2*3125+32)/3 & (3125-32)/3 \\ (2*3125-64)/3 & (3125+64)/3 \end{bmatrix} \\ &= \begin{bmatrix} 2094 & 1031 \\ 2062 & 1063 \end{bmatrix} \end{aligned}$$

(Key insight: Without EVD you'd need 4 matrix multiplications. EVD reduces it to diagonal exponentiation.)

5.2 Singular Value Decomposition (SVD)

SVD: The Most Important Matrix Factorization

EVERY matrix A ($m \times n$) has an SVD, unlike eigendecomposition. $A = U \Sigma V^T$ U : $m \times m$ orthogonal (left singular vectors = eigenvectors of AA^T) Σ : $m \times n$ diagonal (singular values $\sigma_i \geq 0$, in decreasing order) V : $n \times n$ orthogonal (right singular vectors = eigenvectors of ATA)

How to Compute SVD Step-by-Step

13. Compute $A^T A$ ($n \times n$, always symmetric positive semidefinite)
14. Find eigenvalues of $A^T A$: call them $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$
15. Singular values: $\sigma_i = \sqrt{\lambda_i}$ (arrange in decreasing order)
16. Find eigenvectors of $A^T A \Rightarrow$ columns of V
17. Compute left singular vectors: $u_i = (1/\sigma_i) A v_i$ (for $\sigma_i > 0$)
18. If $m > r$ (where $r = \text{rank}$): compute remaining u_i as orthonormal basis for null space of A^T

Worked Example 5.2: Find SVD of $A = \begin{bmatrix} 3 & 0 \\ 4 & 0 \\ 0 & 5 \end{bmatrix}$

A is 3×2 . Compute $A^T A$ (2×2):

$$A^T A = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3^2+4^2+0^2 & 0 \\ 0 & 0+0+5^2 \end{bmatrix} = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}$$

Eigenvalues of $A^T A$: $\det(\begin{bmatrix} 25-\lambda & 0 \\ 0 & 25-\lambda \end{bmatrix}) = (25-\lambda)^2 = 0$
 $\Rightarrow \lambda_1 = \lambda_2 = 25$

Singular values: $\sigma_1 = \sigma_2 = \sqrt{25} = 5$

Eigenvectors of $A^T A = 25I$: any orthonormal basis works!

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Left singular vectors: $u_1 = (1/5) A v_1 = (1/5) \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 3/5 \\ 4/5 \\ 0 \end{bmatrix}$

$$u_2 = (1/5) A v_2 = (1/5) \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

u_3 : must be orthogonal to u_1 and $u_2 \Rightarrow u_3 = \begin{bmatrix} 4/5 \\ -3/5 \\ 0 \end{bmatrix}$ (or $-\begin{bmatrix} 4/5 \\ 3/5 \\ 0 \end{bmatrix}$)

SVD: $A = U \Sigma V^T$

$$U = \begin{bmatrix} 3/5 & 0 & 4/5 \\ 4/5 & 0 & -3/5 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix}$$

$$V^T = I$$

Key SVD Facts

Property	Formula
Rank	$\text{rank}(A) = \text{number of nonzero } \sigma_i$
Frobenius norm	$\ A\ _F = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_r^2}$
Spectral norm (L2 matrix norm)	$\ A\ _2 = \sigma_1$ (largest singular value)
Condition number (via SVD)	$\text{cond}(A) = \sigma_1 / \sigma_r$ (σ_r = smallest nonzero SV)

Property	Formula
Pseudo-inverse	$A^+ = V \Sigma^+ U^T$ (Σ^+ : replace nonzero σ_i with $1/\sigma_i$)

5.3 Low-Rank Matrix Approximation

The truncated SVD gives the best rank-k approximation to A in the Frobenius norm:

Rank-k Approximation:

$$A_k = U_k \Sigma_k V_k^T = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_k u_k v_k^T$$

Approximation error (Eckart-Young Theorem):

$$\min_{\text{rank-k } B} ||A - B||_F = ||A - A_k||_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_r^2}$$

EXAM TIP The Eckart-Young Theorem is very important: truncated SVD is the OPTIMAL low-rank approximation. Used in PCA, image compression, and recommendation systems.

HOOR 6: Vector Calculus I – Derivatives, Partial Derivatives, Gradients, Jacobians

6.1 Univariate Differentiation – Essential Rules

Rule	Formula	Example
Power	$d/dx x^n = n x^{(n-1)}$	$d/dx x^3 = 3x^2$
Exponential	$d/dx e^x = e^x$	$d/dx e^{(2x)} = 2e^{(2x)}$ (chain)
Natural log	$d/dx \ln(x) = 1/x$	$d/dx \ln(3x) = 1/x$
Product rule	$(fg)' = f'g + fg'$	$d/dx (x^2 \sin x) = 2x \sin x + x^2 \cos x$
Quotient rule	$(f/g)' = (f'g - fg')/g^2$	$d/dx (\sin x / x) = (x \cos x - \sin x)/x^2$
Chain rule	$d/dx f(g(x)) = f'(g(x)) \cdot g'(x)$	$d/dx \sin(x^2) = \cos(x^2) \cdot 2x$
Sigmoid	$\sigma'(x) = \sigma(x)(1 - \sigma(x))$	Used in neural networks
tanh	$d/dx \tanh(x) = 1 - \tanh^2(x) = \text{sech}^2(x)$	Used in neural networks

6.2 Partial Derivatives

For $f(x_1, x_2, \dots, x_n)$: the partial derivative $\partial f / \partial x_i$ is computed by differentiating f with respect to x_i while treating all other variables as constants.

Worked Example 6.1: $f(x,y) = x^2y + 3x^2y^2 + y$. Find partial derivatives.

$$df/dx = 2xy + 3y^2 \quad (\text{treat } y \text{ as constant})$$

$$df/dy = x^2 + 6xy + 1 \quad (\text{treat } x \text{ as constant})$$

At point $(x,y) = (1,2)$:

$$df/dx|_{(1,2)} = 2(1)(2) + 3(4) = 4 + 12 = 16$$

$$df/dy|_{(1,2)} = 1 + 6(1)(2) + 1 = 14$$

6.3 The Gradient Vector

Gradient: Direction of Steepest Ascent

For $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (scalar function), the gradient is the column vector of all partial derivatives:
 $\nabla f(x) = [df/dx_1, df/dx_2, \dots, df/dx_n]^T$ The gradient points in the direction where f increases the fastest.

Worked Example 6.2: Gradient of $f(x,y,z) = x^2 + 2xy + z$

$$df/dx = 2x + 2y$$

$$df/dy = 2x$$

$$df/dz = 1$$

$\text{nabla} f = [2x+2y, 2x, 1]^T$
At $(1,1,0)$: $\text{nabla} f = [4, 2, 1]^T$
Direction of steepest ascent at $(1,1,0)$ is $[4, 2, 1]$.
Rate of steepest ascent $= [4,2,1] = \text{sqrt}(16+4+1) = \text{sqrt}(21)$

6.4 Jacobian Matrix

For a vector-valued function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian is the matrix of all first-order partial derivatives:

Jacobian J ($m \times n$ matrix):

$J_{ij} = df_i/dx_j$

The Jacobian generalizes the gradient: it maps small changes in x to small changes in f .

Worked Example 6.3: Find the Jacobian of $f(x,y) = [x^2+y, x*y]^T$
$f_1(x,y) = x^2 + y \Rightarrow df_1/dx = 2x, \quad df_1/dy = 1$ $f_2(x,y) = x*y \Rightarrow df_2/dx = y, \quad df_2/dy = x$
$J = \begin{bmatrix} df_1/dx & df_1/dy \\ df_2/dx & df_2/dy \end{bmatrix} = \begin{bmatrix} 2x & 1 \\ y & x \end{bmatrix}$
At $(x,y) = (2,3)$:
$J = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$

6.5 Key Gradient Identities for ML

Expression $f(x)$	Gradient ∇f	Notes
$a^T x$ (dot product)	a	a is constant vector
$x^T a$ (dot product)	a	same as above
$ x _2^2 = x^T x$	$2x$	Used in L2 regularization
$x^T A x$ (A symmetric)	$2Ax$	Quadratic form; used in optimization
$x^T A x$ (A general)	$(A+A^T)x$	If A not symmetric
$b^T A x$	$A^T b$	Used in least squares
$\text{tr}(X)$ (matrix)	I	Gradient of trace
$\text{tr}(AX)$ (matrix)	A^T	Common in matrix calculus
$\ln \det(X)$ (matrix)	$X^{-T} = (X^{-1})^T$	Used in maximum likelihood

COMMON MISTAKE The gradient of $a^T x$ is a (a column vector), NOT a^T (a row vector). Keep track of vector vs matrix layouts in your computation.

HOUR 7: Vector Calculus II – Matrix Gradients & Backpropagation

7.1 Gradient of Matrix Functions

When $f(X)$ is a scalar function of a matrix X , the gradient dL/dX is a matrix of the same shape as X , where entry (i,j) is $\partial f / \partial X_{ij}$.

Key Matrix Derivative Identities

$f(X)$	df/dX	Shape
$\text{tr}(X^T A)$	A	Same as X
$\text{tr}(AX^T)$	A	Same as X
$\text{tr}(AX)$	A^T	Same as X^T
$\text{tr}(XA)$	A^T	Same as X
$\ AX - b\ _2^2$	$2A^T(AX - b)$	Same as X
$\det(X)$	$\det(X) \cdot X^{-T}$	Same as X

7.2 Chain Rule in Multiple Dimensions

If $y = f(g(x))$ where $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f: \mathbb{R}^m \rightarrow \mathbb{R}^p$:

Jacobian Chain Rule:

$$J_{\{y, x\}} = J_{\{y, g\}} \cdot J_{\{g, x\}} \quad (\text{matrix multiplication of Jacobians})$$

For scalar loss L and intermediate variable $z = f(x)$:

Scalar Chain Rule:

$$dL/dx = (dz/dx)^T \cdot (dL/dz) \quad [\text{note the transpose!}]$$

7.3 Backpropagation in Neural Networks

Backpropagation is the systematic application of the chain rule to compute gradients of a loss function L with respect to all parameters in a neural network.

Forward Pass

For a network with L layers:

$$\begin{aligned}
 z^{(l)} &= W^{(l)} * a^{(l-1)} + b^{(l)} && [\text{pre-activation}] \\
 a^{(l)} &= \text{sigma}(z^{(l)}) && [\text{post-activation, sigma = activation function}] \\
 \text{Loss } L &\text{ is computed from } a^{(L)} \text{ (output) and true labels.}
 \end{aligned}$$

Backward Pass (Key Equations)

Output layer error delta:

$$\delta^{(L)} = \nabla_a L \odot \sigma'(z^{(L)})$$

Hidden layer error (propagate backwards):

$$\delta^{(l)} = (W^{(l+1)})^T \delta^{(l+1)} \odot \sigma'(z^{(l)})$$

Gradient with respect to weights:

$$\nabla_{\{W^{(1)}\}} L = \delta^{(1)} (a^{(1-1)})^T$$

Gradient with respect to biases:

$$\nabla_{\{b^{(1)}\}} L = \delta^{(1)}$$

Here $\odot$ denotes element-wise (Hadamard) product.

Worked Example 7.1: Backprop gradient for linear regression loss
Model: $y_{\text{hat}} = Wx + b$ (single layer, no activation)
Loss: $L = (1/2) y - y_{\text{hat}} ^2 = (1/2) y - Wx - b ^2$
STEP 1: Compute dL/dy_{hat}
Let $r = y_{\text{hat}} - y$ (residual)
$dL/dy_{\text{hat}} = r = (Wx + b - y)$
STEP 2: Chain rule for dL/dW
$dL/dW = dL/dy_{\text{hat}} * d(y_{\text{hat}})/dW$
$d(y_{\text{hat}})/dW = x$ (Jacobian of Wx w.r.t. W is x)
$\Rightarrow dL/dW = r * x^T = (Wx + b - y) * x^T$
STEP 3: dL/db
$d(y_{\text{hat}})/db = 1 \Rightarrow dL/db = r = (Wx + b - y)$
$\Rightarrow$ Update: $W \leftarrow W - \eta * r * x^T, \quad b \leftarrow b - \eta * r$

7.4 Common Gradient Computation Patterns

Least Squares: $\min ||Ax - b||^2$

Gradient:

$$\nabla_x ||Ax - b||_2^2 = 2A^T (Ax - b)$$

Setting gradient to zero (Normal Equations):

$$A^T A x = A^T b$$

Solution (if $A^T A$ invertible):

$$x^* = (A^T A)^{-1} A^T b$$

EXAM TIP The least squares solution $x^* = (A^T A)^{-1} A^T b$ is one of the most important formulas in ML. It gives the best-fit parameters for linear regression.

HOUR 8: Vector Calculus III – Hessian, Taylor Series & Unconstrained Optimization

8.1 The Hessian Matrix

Hessian: Second-Order Information

For $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian $H(x)$ is the $n \times n$ matrix of second partial derivatives: $H_{ij} = d^2f / (dx_i dx_j)$ The Hessian measures the curvature of f at point x .

Hessian definition:

$H = d^2f/dx \, dx^T$ [$n \times n$ symmetric matrix, by Schwarz's theorem]

Hessian for Common Functions

$f(x)$	Gradient ∇f	Hessian H
$a^T x$	a	0 (zero matrix)
$x^T x = \ x\ ^2$	$2x$	$2I$
$x^T A x$ (A symmetric)	$2Ax$	$2A$
$(1/2)\ Ax-b\ ^2$	$A^T(Ax-b)$	$A^T A$

Worked Example 8.1: Find Hessian of $f(x,y) = x^3 + x^2y - 2xy + 5$

First partial derivatives:

$$df/dx = 3x^2 + 2xy - 2y$$

$$df/dy = x^2 - 2x$$

Second partial derivatives:

$$d^2f/dx^2 = 6x + 2y$$

$$d^2f/dy^2 = 0$$

$$d^2f/(dxdy) = 2x - 2 \quad (\text{mixed partial})$$

$$\text{Hessian: } H = \begin{bmatrix} 6x+2y & 2x-2 \\ 2x-2 & 0 \end{bmatrix}$$

$$\text{At point } (1,1): H = \begin{bmatrix} 8 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{At point } (0,0): H = \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}$$

8.2 Definiteness of the Hessian

H is...	Condition	Interpretation at stationary pt
Positive Definite (PD)	All eigenvalues > 0 (or all leading minors > 0)	Strict local MINIMUM

H is...	Condition	Interpretation at stationary pt
Negative Definite (ND)	All eigenvalues < 0	Strict local MAXIMUM
Positive Semidefinite (PSD)	All eigenvalues ≥ 0	Non-strict min (could be flat)
Indefinite	Some $\lambda > 0$, some $\lambda < 0$	SADDLE POINT

Quick PD Check for 2x2 Matrix

Sylvester's Criterion:

$H = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$ is PD iff $a > 0$ AND $ad - b^2 > 0$

8.3 Multivariate Taylor Series

Taylor series lets us approximate f near a point x_0 using derivatives:

First-Order (Linear) Approximation

$$f(x_0 + \delta) \approx f(x_0) + (\nabla f(x_0))^T \delta$$

This is a local linear model of f around x_0 . Used in gradient descent derivation.

Second-Order (Quadratic) Approximation

$$f(x_0 + \delta) \approx f(x_0) + (\nabla f(x_0))^T \delta + (1/2) \delta^T H(x_0) \delta$$

The quadratic term $(1/2)\delta^T H \delta$ captures curvature. This is the foundation for Newton's method.

Worked Example 8.2: Second-order Taylor expansion of $f(x,y) = e^{(x+y)}$ at $(0,0)$

$$f(0,0) = e^0 = 1$$

$$\text{Gradient: } df/dx = e^{(x+y)}, \quad df/dy = e^{(x+y)}$$

$$\text{At } (0,0): \text{ nabla } f = [1, 1]^T$$

$$\text{Hessian: } H = \begin{bmatrix} e^{(x+y)} & e^{(x+y)} \\ e^{(x+y)} & e^{(x+y)} \end{bmatrix}$$

$$\text{At } (0,0): H = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Taylor approximation around $(0,0)$ with $\text{delta} = [dx, dy]^T$:

$$\begin{aligned} f(dx, dy) &\approx 1 + [1, 1] \cdot [dx, dy]^T + (1/2) \cdot [dx, dy] \cdot \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot [dx, dy]^T \\ &= 1 + dx + dy + (1/2) \cdot (dx^2 + 2 \cdot dx \cdot dy + dy^2) \\ &= 1 + dx + dy + (1/2) \cdot (dx + dy)^2 \end{aligned}$$

8.4 Unconstrained Optimization

We want to find x^* that minimizes (or maximizes) a differentiable function $f(x)$.

First-Order Necessary Condition (FONC)

$$\nabla f(x^*) = 0 \quad (x^* \text{ is a stationary point})$$

Every local min/max satisfies FONC, but NOT every stationary point is a min/max (could be saddle).

Second-Order Sufficient Conditions (SOSC)

$H(x^*)$ is PD $\Rightarrow x^*$ is a strict local minimum
 $H(x^*)$ is ND $\Rightarrow x^*$ is a strict local maximum
 $H(x^*)$ is indefinite $\Rightarrow x^*$ is a saddle point

Problem-Solving Steps: Find & Classify Critical Points

19. Compute the gradient $\nabla f(x)$
20. Solve $\nabla f(x) = 0$ to find all stationary points x^*
21. Compute the Hessian $H(x)$
22. Evaluate H at each stationary point
23. Classify: PD $\Rightarrow$ minimum, ND $\Rightarrow$ maximum, indefinite $\Rightarrow$ saddle

Worked Example 8.3: Find and classify critical points of $f(x,y) = x^2 + y^2 - 2x - 4y + 5$

STEP 1: Gradient

$$df/dx = 2x - 2$$

$$df/dy = 2y - 4$$

STEP 2: Set gradient = 0

$$2x - 2 = 0 \Rightarrow x = 1$$

$$2y - 4 = 0 \Rightarrow y = 2$$

Critical point: (1, 2)

STEP 3 & 4: Hessian

$$H = \begin{bmatrix} d^2f/dx^2 & d^2f/dxdy \\ d^2f/dydx & d^2f/dy^2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

STEP 5: Classify

Eigenvalues of $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$: both equal $2 > 0 \Rightarrow H$ is PD

Sylvester: $a=2>0$, $\det=4>0 \checkmark$

$\Rightarrow (1,2)$ is a strict local (global) MINIMUM

$$f(1,2) = 1 + 4 - 2 - 8 + 5 = 0$$

Worked Example 8.4: Find critical points of $f(x,y) = x^2 - y^2$

Gradient: $df/dx = 2x$, $df/dy = -2y$

Setting to 0: $x=0$, $y=0 \Rightarrow$ critical point (0,0)

Hessian: $H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$

Eigenvalues: 2 and -2 $\Rightarrow H$ is INDEFINITE

$\Rightarrow (0,0)$ is a SADDLE POINT (function increases in x-direction, decreases in y-direction)

8.5 Convexity

Convexity is a crucial concept in optimization. Convex problems have no local minima that are not global.

Convexity Definition:

f is convex iff $f(\lambda u + (1-\lambda)v) \leq \lambda f(u) + (1-\lambda)f(v)$ for all u, v and $\lambda \in [0, 1]$

Differentiability Criterion:

f is convex iff $H(x)$ is PSD (positive semidefinite) for all x

Property	Convex f	Strict Convex f
Local min = global min	Yes	Yes (unique global min)
Hessian	H is PSD everywhere	H is PD everywhere
Examples	x^2 , $\ x\ $, e^x	x^2 , e^x (strictly)
Gradient descent	Guaranteed to converge to global min	Converges to unique minimum

EXAM TIP If the Hessian $H = A^T A$ (as in least squares), it is always PSD, making least squares a convex problem. This guarantees a global minimum.

8.6 Gradient Descent (Preview)

Gradient descent iteratively moves in the direction of steepest descent to minimize f :

Update rule:

$x_{t+1} = x_t - \eta \nabla f(x_t)$

η (eta) is the learning rate (step size).

Choice of eta	Effect
eta too large	>Overshooting minimum; oscillation; possible divergence
eta just right	Smooth convergence to minimum
eta too small	Very slow convergence (many iterations needed)

NOTE Full treatment of gradient descent variants (SGD, Momentum, Adam, etc.) is in Hours 9-11 which is outside the mid-sem scope. For the mid-sem, understand the basic update rule, its derivation from Taylor series, and convergence condition.

MASTER FORMULA CHEAT SHEET

Linear Algebra

Topic	Formula	When to Use
Solve $Ax=b$	Row reduce $[A b]$; or $x=A^{-1}b$ if A invertible	Any linear system
Consistency	$\text{rank}(A) = \text{rank}([A b])$	Check if solution exists
2x2 inverse	$A^{-1} = (1/\det) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$	Quick inverse
Rank-Nullity	$\text{rank}(A) + \text{nullity}(A) = n$	Dimension calculations
Subspace check	$0 \in S$; closed under $+$ and $\cdot$	Verify subspace
L2 norm	$\ x\ = (x^T x)^{1/2}$	Length of vector
Inner product	$\langle x, y \rangle = x^T y$	Angle, projection
Projection	$\text{proj}_y(x) = (\langle x, y \rangle / \langle y, y \rangle) y$	Decompose vectors
Gram-Schmidt	$v'_k = v_k - \sum \langle v_k, u_j \rangle u_j$, normalize	Orthonormal basis
Char equation	$\det(A - \lambda I) = 0$	Find eigenvalues
Trace/Det	$\text{tr} = \sum(\lambda)$, $\det = \prod(\lambda)$	Verify eigenvalues
EVD	$A = PDP^{-1}$	Diagonalize, powers
SVD	$A = U \Sigma V^T$	Any matrix, best approx
Rank-k approx	$A_k = U_k \Sigma_k V_k^T$	Best low-rank approx
Cholesky	$A = LL^T$ (SPD only)	Efficient solve

Calculus & Optimization

Topic	Formula	Notes
Gradient	$[df/dx_1, \dots, df/dx_n]^T$	Column vector
Jacobian	$J_{ij} = df_i/dx_j$ ($m \times n$)	Vector-valued f
Hessian	$H_{ij} = d^2 f / (dx_i dx_j)$	$n \times n$ symmetric
Taylor 1st	$f(x+d) \approx f(x) + (\text{nabla} f)^T d$	Linear approx
Taylor 2nd	$f(x+d) \approx f(x) + (\text{nabla} f)^T d + (1/2) d^T H d$	Quadratic approx
Grad of $a^T x$	a	Constant vector
Grad of $\ x\ ^2$	$2x$	Regularization
Grad of $x^T A x$	$2Ax$ (symmetric A)	Quadratic form
Grad of $\ Ax-b\ ^2$	$2A^T(Ax-b)$	Least squares
Normal equations	$A^T A x = A^T b$	Least squares solution
Stationary pt	$\text{nabla} f(x^*) = 0$	Necessary condition
Min condition	$H(x^*)$ is PD	Sufficient condition
Convex test	H PSD everywhere	Global min guaranteed
Gradient descent	$x_{t+1} = x_t - \eta \cdot \text{nabla} f$	Iterative optimization

Common Exam Pitfalls to Avoid

- NOT checking consistency (rank test) before solving $Ax=b$
- Using A^{-1} when A is singular ($\det=0$): always check $\det$ first
- Confusing column space (use original pivot columns) vs row space (use REF rows)
- Forgetting factor of 2 in grad of $\|x\|^2$: answer is $2x$, not x
- Missing the $(1/2)$ factor in Taylor 2nd order approximation
- Applying EVD to a non-diagonalizable or rectangular matrix: use SVD instead
- Not verifying eigenvalues with $\text{tr}(A)=\text{sum}$ and $\det(A)=\text{product}$
- Not normalizing eigenvectors when computing orthonormal basis
- Confusing H PD (min) vs H ND (max) vs indefinite (saddle)
- Computing gradient as a row vector instead of a column vector
- Forgetting that for a symmetric matrix A , EVD gives $A = Q D Q^T$ where $Q^{-1}=Q^T$
- In Gram-Schmidt: not computing the orthogonal component before normalizing

All the Best for Your Exam! — AIMLC ZC416